ENGINEERING ECONOMICS

LECTURE - 09

Effective Annual Interest Rate

- Example:
  - “***12% annual rate, compounded monthly***”
- Pick this statement apart:
  - **12% is the nominal interest rate**
  - **“Compounded monthly” tells us the number of compounding periods in a year (12)**

Effective Annual Interest Rate

- Example:
- The effective interest rate per month is 1%:
  - **We would like to be able to convert this to an effective annual interest rate**

Effective Annual Interest Rate

The effective annual interest rate i for a nominal interest rate r compounded m times per year is:

$$i = (1 + r / m)^m - 1$$

Monthly Compounding Example

- Given:

$r = 9\%$ per year, compounded monthly

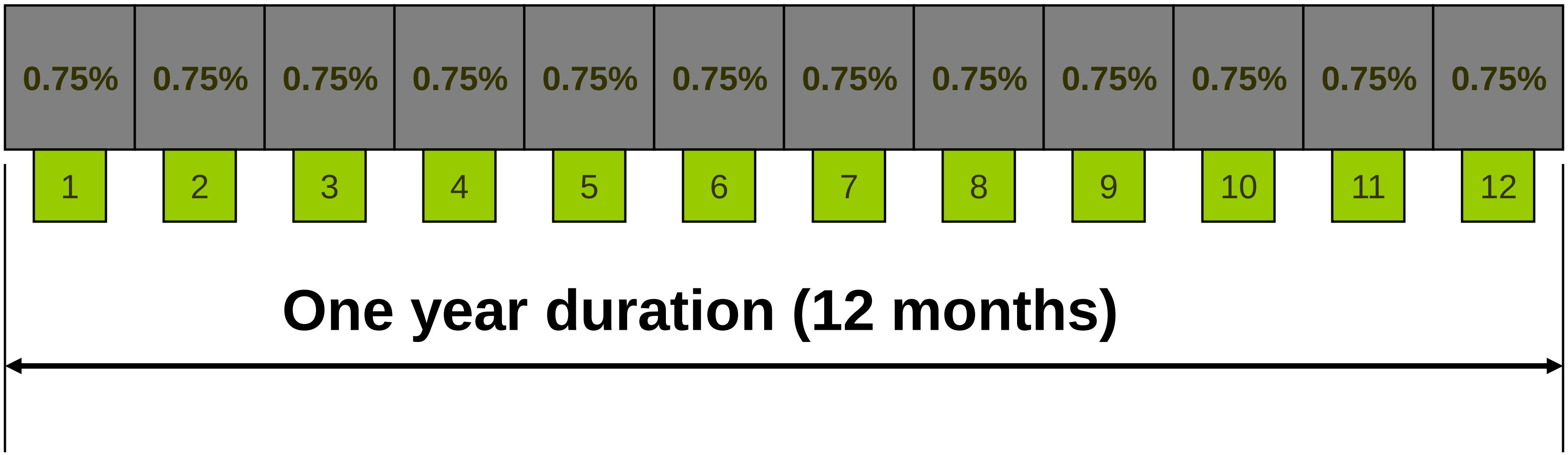
Compounding is monthly, so there are $m = 12$ compounding periods in a year

Effective monthly rate:
$0.09/12 = 0.0075 = 0.75\%$/month

Effective annual rate:
$(1 + 0.0075)^{12} - 1 = 0.0938 = 9.38\%$/year

Example (continued)

- r = 9% is the nominal rate
- "Compounded monthly" means m = 12
- The effective monthly rate is 0.75%/month
- The effective annual rate is 9.38% per year



Quarterly Compounding Example

- Given r = 9% per year, compounded quarterly

Quarter 1	Quarter 2	Quarter 3	Quarter 4

What is the effective rate?

- **0.09/4 = 0.0225 = 2.25%/quarter is the effective quarterly rate**
- **$(1 + .0225)^4 - 1 = 0.0930$ = 9.30%/year is the effective annual rate**

Weekly Compounding Example

- **Given r = 9% per year, compounded weekly:**
  - **Assume 52 weeks per year**

  - **The effective weekly rate is (0.09/52) = 0.00173 = 0.173%/week**

  - **The effective annual rate is $(1 + 0.00173)^{52} - 1$ = 0.0940 = 9.40%/year**

Comparison

- The effective annual interest rate is always greater than the nominal interest rate:
  - You are earning (paying) interest on your interest

- The difference is greater with more frequent compounding:
  - If compounded *quarterly*, we get 9.30%/year
  - If compounded *monthly*, we get 9.38%/year
  - If compounded *weekly*, we get 9.40%/year

- What if we compound infinitely often?

Effective Interest Rate per Payment Period (*i*)

$$i = [1 + r / CK]^C - 1$$

C = number of interest periods per payment period

K = number of payment periods per year

CK = total number of interest periods per year, or M

r/K = nominal interest rate per payment period

Case 1: 8% compounded quarterly

Payment Period = Quarter

Interest Period = Quarterly

1st Q	2nd Q	3rd Q	4th Q

1 interest period

Given $r = 8\%$,

$K = 4$ payments per year

$C = 1$ interest period per quarter

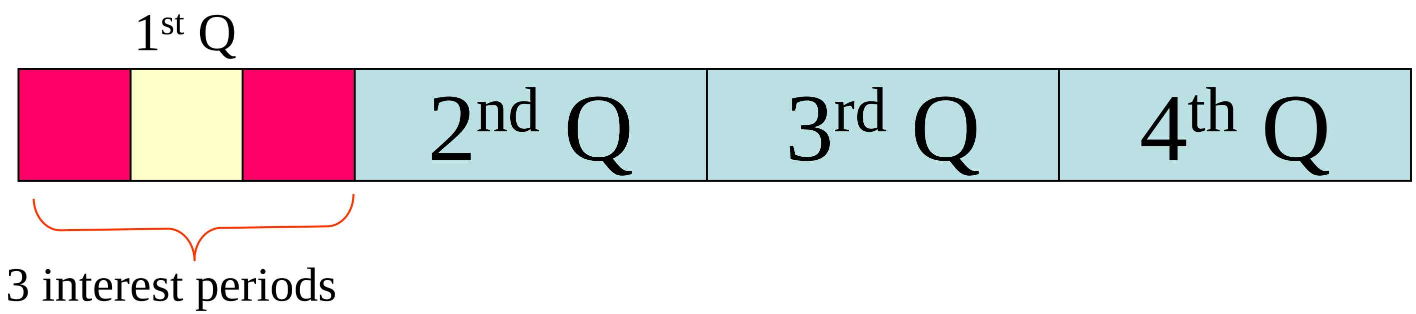
$M = 4$ interest periods per year

$$
\begin{aligned}
i &= [1 + r/CK]^{C} - 1 \\
&= [1 + 0.08/(1)(4)]^{1} - 1 \\
&= 2.000\% \text{ per quarter}
\end{aligned}
$$

Case 2: 8% compounded monthly

Payment Period = Quarter

Interest Period = Monthly



Given r = 8%,

K = 4 payments per year

C = 3 interest periods per quarter

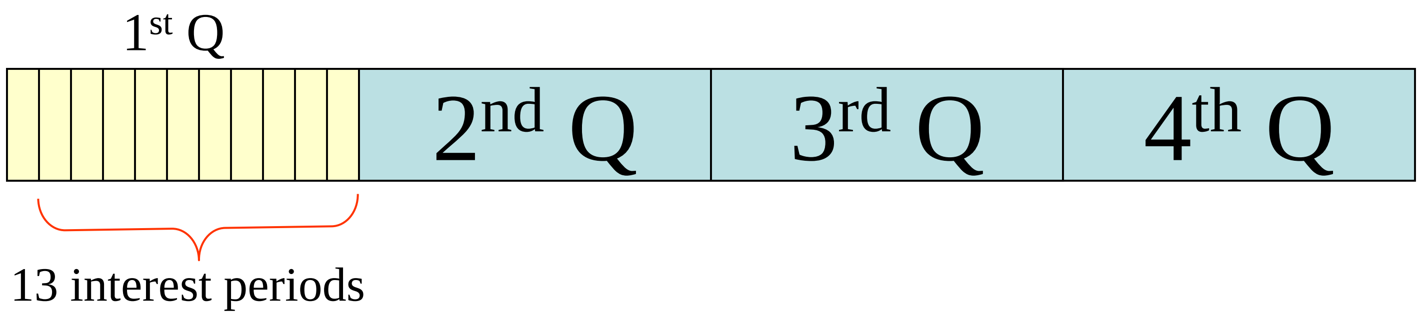
M = 12 interest periods per year

$$\begin{aligned} i &= [1 + r / CK]^{C} - 1 \\ &= [1 + 0.08 / (3)(4)]^{3} - 1 \\ &= 2.013\% \text{ per quarter} \end{aligned}$$

Case 2: 8% compounded weekly

Payment Period = Quarter

Interest Period = Weekly



Given r = 8%,

K = 4 payments per year

C = 13 interest periods per quarter

M = 52 interest periods per year

$$\begin{aligned} i &= [1 + r / CK]^{C} - 1 \\ &= [1 + 0.08 / (13)(4)]^{13} - 1 \\ &= 2.0186\% \text{ per quarter} \end{aligned}$$

Effective Interest Rate per Payment Period with Continuous Compounding

$$\boxed{i = [1 + r / CK]^{C} - 1}$$

where CK = number of compounding periods per year

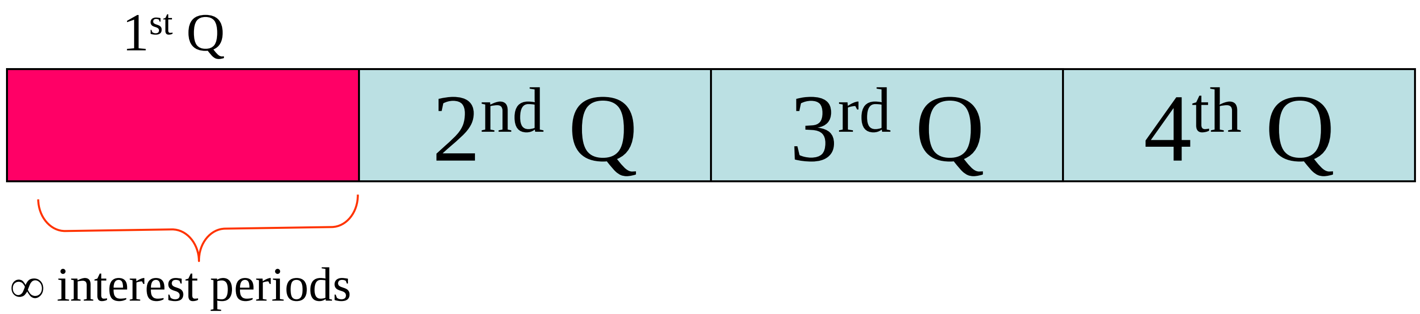
continuous compounding => $C \rightarrow \infty$

$$i = \lim[(1 + r / CK)^{C} - 1]$$
$$= (e^{r})^{1/K} - 1$$

Case 3: 8% compounded continuously

Payment Period = Quarter

Interest Period = Continuously



Given r = 8%,

K = 4 payments per year

$$i = e^{r/K} - 1$$

$$= e^{0.02} - 1$$

$$= 2.0201\% \text{ per quarter}$$

Summary: Effective interest rate per quarter

Case 0	Case 1	Case 2	Case 3
8% compounded quarterly	8% compounded monthly	8% compounded weekly	8% compounded continuously
Payments occur quarterly	Payments occur quarterly	Payments occur quarterly	Payments occur quarterly
2.000% per quarter	2.013% per quarter	2.0186% per quarter	2.0201% per quarter

Which One to Use: r or i

- Some problems state only the nominal interest rate:
  - The *nominal interest rate* is frequently stated for loans
- The effective interest rate is always the one used in:
  - Published interest tables
  - time-value-of-money formulas
  - Spreadsheet functions
- Remember:
  - Always use the effective interest rate in solving problems
  - (Either annual or per period)

Discussion